HARDWARE SECURITY

Agenda

- Physical Hardware Security

Threat vectors

- Interception
  - Gain access to information without interfering with system
- Interruption
  - Prevention of system functionality
- Modification
  - Invasive tampering
- Fabrication
- Counterfeiting

Security Categories

- Physical
- Logical
- Environmental
- Operational

Physical attacks

- Invasive
  - Direct access to embedded components (e.g. cpu)
    - Micro probing, reverse engineering, memory readout techniques (e.g. freezing)
    - Require lot of time, knowledge and resources
- Semi-invasive (integrated chip cards)
  - UV lights, x-rays, laser, EM field, heating
  - Optical fault induction (SRAM illumination)
  - Low cost, easy reproduction on same target

Logical Security

- Access Control
- Cryptographic Algorithms
- Cryptographic Protocols

Logical attacks

- Software Attacks – API
  - No specialised equipment needed
  - Very fast
- Issues:
  - Integrity of keys
  - Function parameter checking
  - Security policy enforcement

Environmental security

- Device itself is the asset
- Goal – limit attacker's possibilities by creating layers of hindrance (e.g. access)
- Administrative controls should be part of security policy

Operational security

- Security risks related to operation of hardware
- Example: ATMs
- User's knowledge of:
  - Real vs. Fake card reader
  - Keypad operation
  - PIN Safeguarding
  - Latest attacks

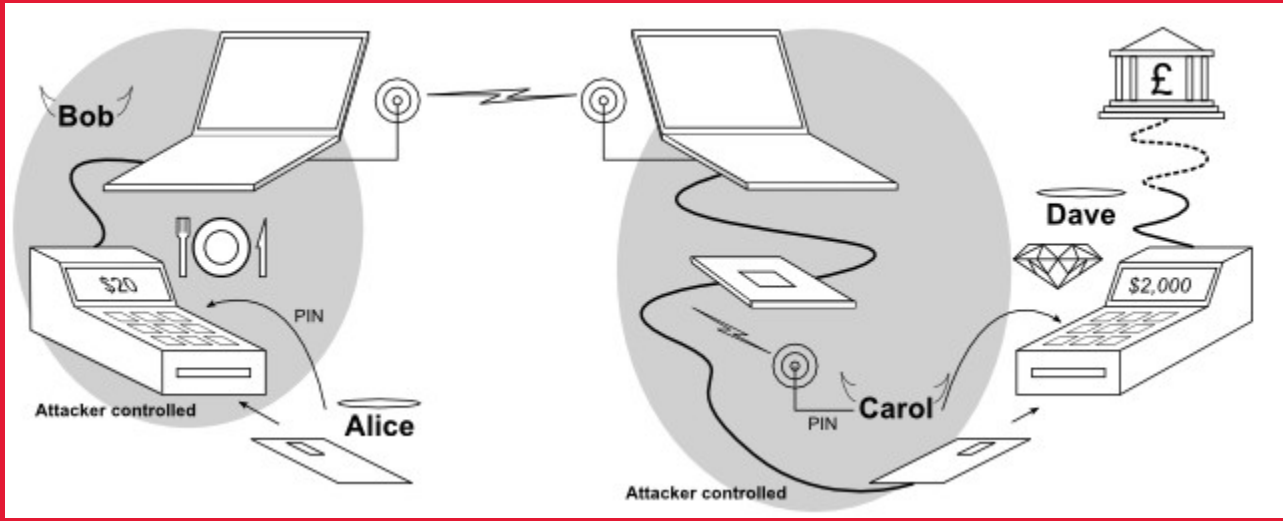
Hardware security modules

- For secure generation and storage of crypto information
  - Often physically tamper resistant
  - Sometimes have H/W cryptographic acceleration
  - Sometimes have special “trusted” peripherals (e.g. card readers, key pads, etc..)
  - Example: Banks
    - ATMs
    - Pre-payment electricity meters

Examples

- Credit Cards
  - Magnetic Stripes
  - Chip & PIN
  - RFID (Radio Frequency Identification)

Example: Chip and Pin Relay Attack



Goals

- Learn the concepts of designing secure hardware
- Become familiar with types of attacks and attackers

General Security Concepts

- Nothing is ever 100% secure
  - Given enough time, resources, and motivation, an attacker can break any system
- Secure your product against a specific threat
  - What needs to be protected
  - Why it is being protected
  - Who you are protecting against (define the enemy)

Security During Product Development

- Establish a security policy as the "foundation" for design
- Treat security as an integral part of your product's development
- Minimize the elements you need to secure
- Reduce risk to an acceptable level
  - Elimination of all risk is not cost-effective

Security During Product Development (cont)

- Implement layered security
- Do not implement unnecessary security mechanisms
  - Each mechanism should support a defined goal
- Costs of a successful attack should outweigh potential rewards

Types of Attack

- Insider Attack
  - Significant percentage of breaches
  - Ex.: Run-on fraud, disgruntled employees
- Lunchtime Attack
  - Take place during a small window of opportunity
  - Ex.: During a lunch or coffee break
- Focused Attack
  - Time, money, and resources not an issue

Types of attack (cont)

- Invasive Attacks
  - Hardware Reverse Engineering
- Semi-invasive Attacks
  - Heating
- Non-Invasive Attacks
  - E.M. Radiation

Types of Attackers

- Clever Outsiders
  - Intelligent, but have limited knowledge of the product
  - Usually take advantage of a known weakness
  - Ex.: Curious kids, college students
- Knowledgeable Insiders
  - Substantial specialized technical experience
  - Highly sophisticated tools and instruments
  - Ex.: Professional engineers

Types of Attackers (cont)

- Funded Organizations
  - Specialists backed by great funding resources
  - In-depth analysis, sophisticated attacks, most advanced analysis tools
  - Ex.: Government, organized crime

Accessing the Product

- Purchase
  - Attacker buys the product from a retail store
- Evaluation
  - Attacker rents or borrows the product
- Active
  - Product is in operation, not owned by attacker
- Remote Access
  - No physical access to product
  - Attacks launched remotely

Threat Vectors

- Interception (or Eavesdropping)
  - Gain access to information without opening the product
- Interruption (or Fault Generation)
  - Preventing the product from functioning normally
- Modification
  - Invasive tampering of the product
- Fabrication
  - Creating counterfeit data in a product

Goals of an Attack

- Competition (or Cloning)
  - Specific theft to gain marketplace advantage
- Theft-of-Service
  - Obtaining a service for free that normally costs money
- User Authentication (or Spoofing)
  - Forging a user's identity to gain system access
- Privilege Escalation (or Feature Unlocking)
  - Gaining increased command of a system or unlocking hidden/undocumented features

Physical security

- Tampering
  - “An (physical) interference of a harmful nature”
  - Tamper Mechanisms:
    - Strive to prevent an attempt by an attacker to perform unauthorised physical or electronic action

Tamper mechanisms

- Tampering
  - *"An (physical) interference of a harmful nature"*
  - Tamper Mechanisms:
    - Strive to prevent an attempt by an attacker to perform unauthorised physical or electronic action

Anti-Tamper Mechanisms

- Primary area of physical security for embedded systems
- Attempts to prevent unauthorized physical or electronic tampering against the product
- Most effectively used in layers
- Possibly bypassed with knowledge of method
  - Attackers may intentionally destroy a device to determine its security mechanisms

Anti-Tamper Mechanisms (cont)

- Tamper Resistance
  - Specialized materials used to make tampering difficult
  - Ex.: One-way screws, epoxy encapsulation
- Tamper Evidence
  - Ensure that there is visible evidence left behind by tampering
  - Only successful if a process is in place to check for deformity
  - Ex.: Passive detectors (seals, tapes, glues), special enclosure finishes ("bleeding paint")

Anti-Tamper Mechanisms (cont)

- Tamper Detection
  - Enable the hardware device to be aware of tampering
  - Switches: Detect the opening of a device or breach of security boundary
  - Sensors: Detect an operational or environmental change (ex.: temperature, voltage, radiation)
  - Circuitry: Detect a puncture, break, or attempted modification of a defined security envelope (ex.: nichrome wire, W.L. Gore's D3 enclosure)

Anti-Tamper Mechanisms (cont)

- Tamper Response
  - Countermeasures taken upon the detection of tampering
  - Ex.: Erase memory, shutdown/disable device, enable logging
- *Physical Security Devices for Computer Subsystems* provides comprehensive attacks and countermeasures

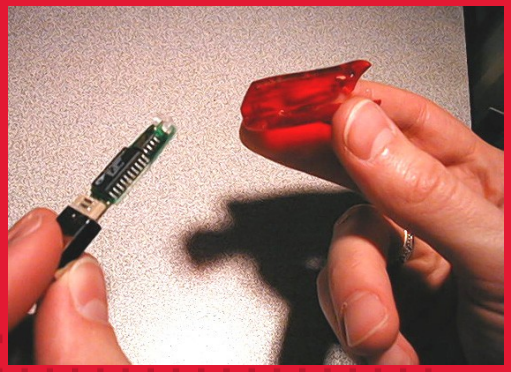
Enclosure & Mechanical

- Product Housing
- External Interfaces

Product Housing

- Attack goal of opening the product is to get access to internal circuitry
- Usually as easy as loosening some screws or prying open the device
- Designers should prevent easy access to product internals

Product Housing (cont)



Product Housing (cont)

- Sealed or molded housing
  - Use a high-melting point glue
  - Use ultrasonic welding to create a one-piece outer shell
  - Will require destruction of device to open it
  - Consider service issues (if a legitimate user can open device, so can attacker)

Product Housing (cont)

- Security bits and one-way screws
  - Prevents housing from being easily opened
  - Ex.: 3.8mm, 4.5mm, and Tri-Wing screw for Nintendo and Sega cartridges/consoles
  - Beware: Attackers can purchase many of these special bits online

External Interfaces

- Usually connects a product to the outside world
  - Manufacturing tests, field programming/upgrading, peripheral connections
  - Ex.: RS232, USB, Firewire, Ethernet, JTAG (IEEE 1149.1)

External Interfaces (cont)

- Will likely be probed or monitored by attacker
- Only publicly known information should be passed
- Encrypt secret or critical components
  - If they must be sent at all...
  - Ex.: Palm OS system password decoding attack

External Interfaces (cont)

- Don't just hide the interface
  - Will easily be discovered by an attacker
  - Ex.: Proprietary connector types, hidden access doors or holes, stickers
- Protect against malformed, bad packets
  - Intentionally sent by attacker to cause fault

External Interfaces (cont)

- Physically remove all diagnostic, debug, and backdoor interfaces from production units
  - Even if they are undocumented
  - Difficult to do
  - Do not just cut traces or remove resistors (which could be repaired by an attacker)
  - Ex.: Intel NetStructure crypto accelerator administrator access, Palm OS debug mode

External Interfaces (cont)

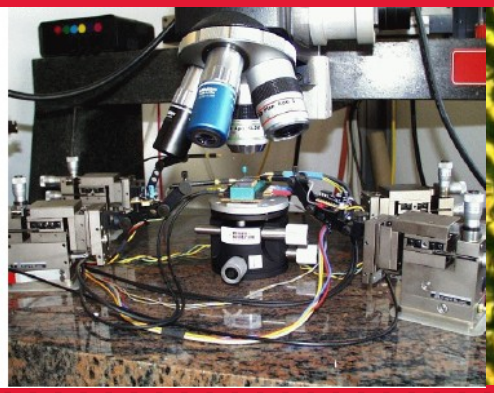
- Field programmability
  - Only allow new versions of firmware to be loaded into product (so attacker can not make use of old, known security flaws)
  - Do not release firmware on your Web site (could be disassembled and analyzed by attacker)
  - If you must, use code signing (DSA) or hashes (SHA-1, MD5) to verify integrity
  - Even better, encrypt firmware images

Circuit Board

- Physical Access to Components
- EMI/ESD/RF Interference
- PCB Design and Routing
- Memory and Programmable Logic
- Power Supply
- Cryptographic Processors and Algorithms

Physical Attacks

- Micro-probing station

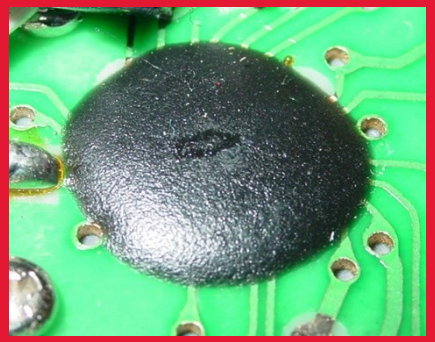


Access to Components

- Giving an attacker easy access to components aids in reverse engineering of the product
- Make sensitive components difficult to access
  - Ex.: Microprocessor, ROM, RAM, ASICs, FPGAs
- Remove identifiers and markings from ICs
  - Use stainless steel brush, small sander, micro-bead blast, laser etcher, or third-party
  - Easy for attacker to find data sheets online

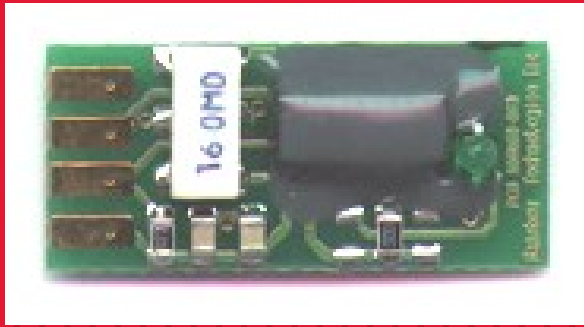
Access to Components (cont)

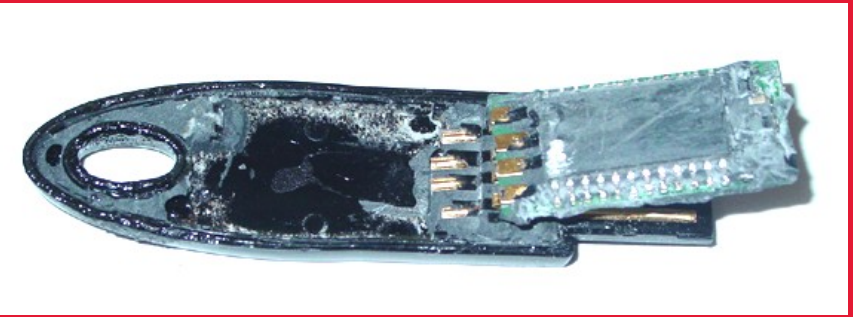
- Use advanced package types
  - Difficult to probe using standard tools
  - Ex.: BGA, Chip-on-Board (COB), Chip-in-Board (CIB)
- Use proprietary or customized ICs



Access to Components (cont)

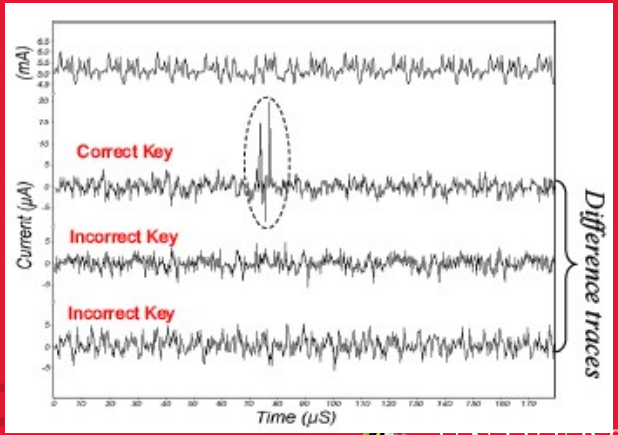
- Cover critical components with epoxy or urethane encapsulation
  - Usually used to protect circuitry from moisture, dust, mold, corrosion, or arcing
  - Difficult, but not impossible, to remove with chemicals or tools





Logical attacks

- Non-Invasive
- No Physical Damage
- Monitoring/Eavesdropping
  - TEMPEST attacks
  - Side Channel Attacks
  - Timing Analysis
  - Power Analysis
  - Fault Analysis



EMI/ESD/RF Interference

- All devices generate electromagnetic interference (EMI)
- Can be monitored and used by attacker to determine secret information
  - Ex.: Data on a computer monitor, cryptographic key from a smartcard
- Devices may also be susceptible to RF or electrostatic discharge (ESD)
  - Intentionally injected to cause failure

EMI/ESD/RF Interference (cont)

- Install EMI shielding
  - Decrease emissions and increase immunity
  - Ex.: Coatings, tapes, sprays, housings
  - Be aware of changes in thermal characteristics that shielding may introduce (heating)
- Prevent against ESD on exposed I/O lines
  - Clamping diodes or Transient Voltage Suppressors
  - Ex.: Keypads, buttons, switches, display
- Keep circuit traces as short as possible

EMI/ESD/RF Interference (cont)

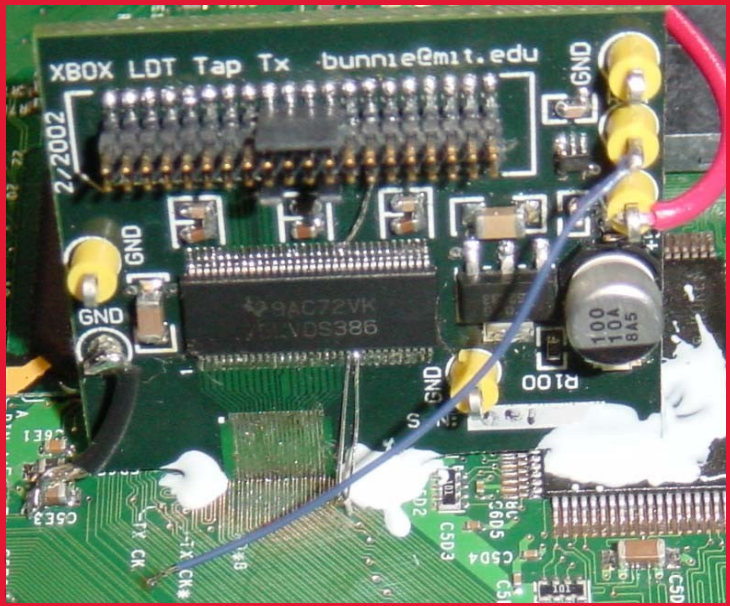
- Use properly designed power and ground planes
- Power supply circuitry as physically close as possible to power input
- Remove unnecessary test points
  - Use filled pad as opposed to through-hole, if necessary
- Unused I/O pins and modules should be disabled or set to fixed state

PCB Design and Routing

- Confuse trace paths to prevent easy reverse engineering
  - Hide critical traces on inner board layers
  - Be aware of data being transferred across exposed and/or accessible address, data, and control buses
- Use buried vias whenever possible
  - Connects between two or more inner layers but no outer layer
  - Cannot be seen from either side of the board

PCB Design and Routing (cont)

- Ex.: Tap board used to intercept data transfer over Xbox's HyperTransport bus



Memory and Programmable Logic

- Most memory is insecure
  - Can be read with standard device programmer
  - Serial EEPROMs can be read in-circuit, usually SPI or I^2C bus (ex.: USB authentication token)
- Difficult to securely and totally erase data from RAM and non-volatile memory
  - Remnants may exist and be retrievable from devices long after power is removed

Memory and Programmable Logic (cont)

- SRAM-based FPGAs most vulnerable to attack
  - Must load configuration from external memory
  - Bit stream can be monitored to retrieve data
- Protect against I/O scan attacks
  - Attacker cycles through all possible combinations of inputs to determine outputs
  - Use unused pins to detect probing

Memory and Programmable Logic (cont)

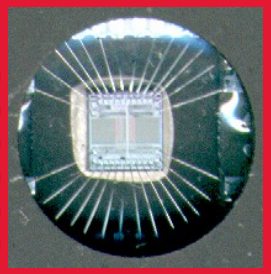
- Security fuses and boot-block protection
  - Enabled for "write-once" access to a memory area or to prevent full read back
  - Implement if available
  - Ex.: PIC16C84 attack in which security bit is removed by increasing VCC during repeated write accesses

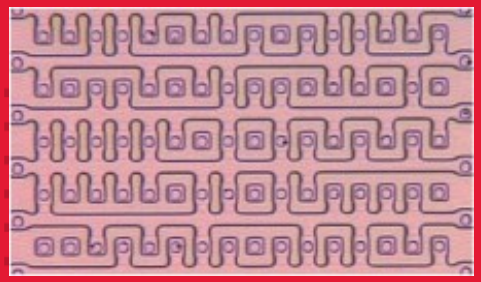
Memory and Programmable Logic (cont)

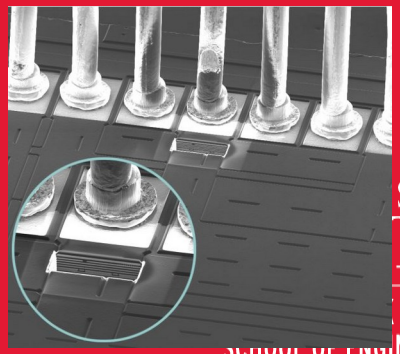
- Limit the amount of time that critical data is stored in the same region of memory
  - Periodically flip the stored bits
- If using state machine, ensure all conditions and defaults are covered
- Add digital "watermarks"
  - Features or attributes in design that can be uniquely identified as being rightfully yours

Memory and Programmable Logic (cont)

- Chip Decapping and Die Analysis attacks
  - Attacker can visually recreate contents or modify die (Ex.: to obtain crypto key or remove security bit)
  - Tools: Chip Decappers, Scanning Electron Microscope, Voltage Contrast Microscopy, Focused Ion Beam







Power Supply

- Define minimum and maximum operating limits
  - Ex.: Comparators, watchdogs, supervisory circuits
- Do not rely on end user to supply a voltage within recommended operating conditions
  - Implement linear regulator or DC-DC converter

Power Supply (cont)

- Simple Power Analysis (SPA)
  - Attacker directly observes power consumption
  - Varies based on microprocessor operation
  - Easy to identify intensive functions (ex.: cryptographic)
- Differential Power Analysis (DPA)
  - Advanced mathematical methods to determine secret information on a device

Cryptographic Processors and Algorithms

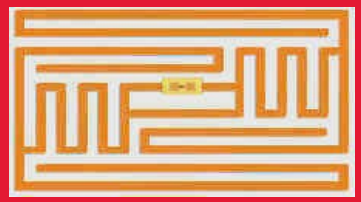
- Strength of cryptography relies on secrecy of key, not the algorithm
  - Do not create your own crypto algorithms
- It is not safe to assume that large key size will guarantee security
- If algorithm implemented improperly, can be broken or bypassed by attacker
  - Test implementations in laboratory first!

Cryptographic Processors and Algorithms (cont)

- Move cryptographic processes out of firmware and into FPGA
  - Harder to probe than ROM devices
  - Increased performance (more efficient)
- Or, use secure cryptographic coprocessor
  - Self-contained, hardware tamper response, authentication, general-purpose processor
  - Ex.: Philips VMS747, IBM 4758

Example: RFID

- RFID – Radio Frequency Identification
  - Originally developed as the “Barcode of the future”
- Now used as
  - Inventory control
  - Logistics and supply chain management
  - Physical access cards
  - Payment
  - Motorway charges
  - Gas stations
  - Small items in shop



Concerns

- Examples:
  - Future:
  - Embedded in all kinds of devices
  - From clothing, to all products we buy
    - e.g. Milk that will tell fridge when it is expired
- Issues:
  - Privacy
  - Security – RFID was not designed with security in mind!!

RFID Concerns

- Attacks which RFID is susceptible to:
  - Susceptible to Power Analysis attacks
  - Can be susceptible to Cloning attacks
  - Susceptible to Relay attacks
    - “Is your cat infected with a computer virus?”

Conclusion

- Determine what to protect, why you are protecting it, and who you are protecting against
  - No one solution fits everyone
- Do not release product with a plan to implement security later
  - It usually never happens...
- Nothing is 100% secure

Conclusion (cont)

- Be aware of latest attack methodologies & trends
- As design is in progress, allocate time to analyze and break product
- Learn from mistakes
  - Study history and previous attacks

References

- PIC Microcontroller Discussion List, "Re: Code protect," Posted April 26, 1995, **www.brouhaha.com/~eric/pic/84security.html**
- J. Grand (Kingpin), "Attacks on and Countermeasures for USB Hardware Token Devices," *Proceedings of the Fifth Nordic Workshop on Secure IT Systems*, 2000, **www.grandideastudio.com/files/security/tokens/usb_hardware_token.pdf**
- P. Gutmann, "Secure Deletion from Magnetic and Solid-State Memory Devices," *Sixth USENIX Security Symposium*, 1996, **www.usenix.org/publications/library/proceedings/sec96/full_papers/gutmann/index.html**
- A. Huang, "Hacking the Xbox: An Introduction to Reverse Engineering," No Starch Press, 2003.
- J.R. Rao and P. Rohatgi, "EMPowering Side-Channel Attacks," IBM Research Center, www.research.ibm.com/intsec/emf-paper.ps
- S.H. Weingart, "Physical Security Devices for Computer Subsystems: A Survey of Attacks and Defenses," *Workshop on Cryptographic Hardware and Embedded Systems*, 2000.
- J. Grand (Kingpin), “Palm OS Password Retrieval and Decoding,” September 2000, **www.grandideastudio.com/files/security/mobile/palm_password_decoding_advisory.txt**
- B. Oblivion, “Intel NetStructure Backdoors,” May 2000, www.atstake.com/research/advisories/2000/ipivot7110.html and ipivot7180.html
- J. Grand (Kingpin), “Palm OS Password Lockout Bypass,” March 2001, **www.grandideastudio.com/files/security/ mobile/palm_backdoor_debug_advisory.txt**
- W. van Eck, “Electronic Radiation from Video Display Units: An Eavesdropping Risk?” *Computers and Security*, 1985, **www.jya.com/emr.pdf**